CN334 Data Analytics Lab

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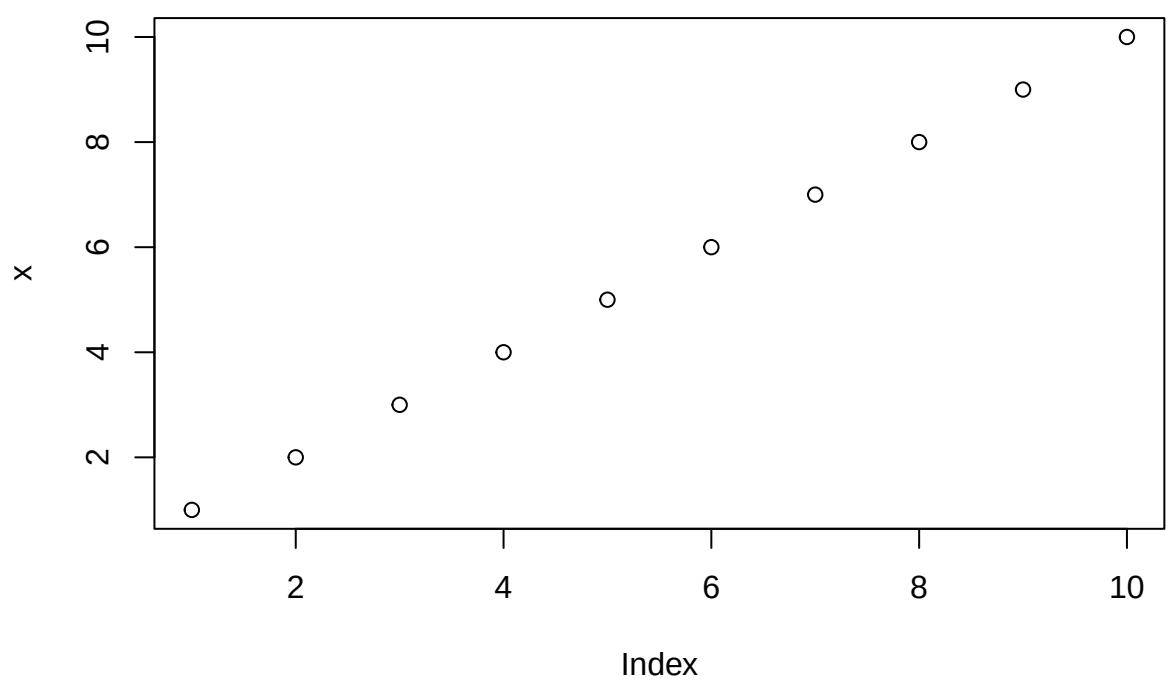
Exercise 5

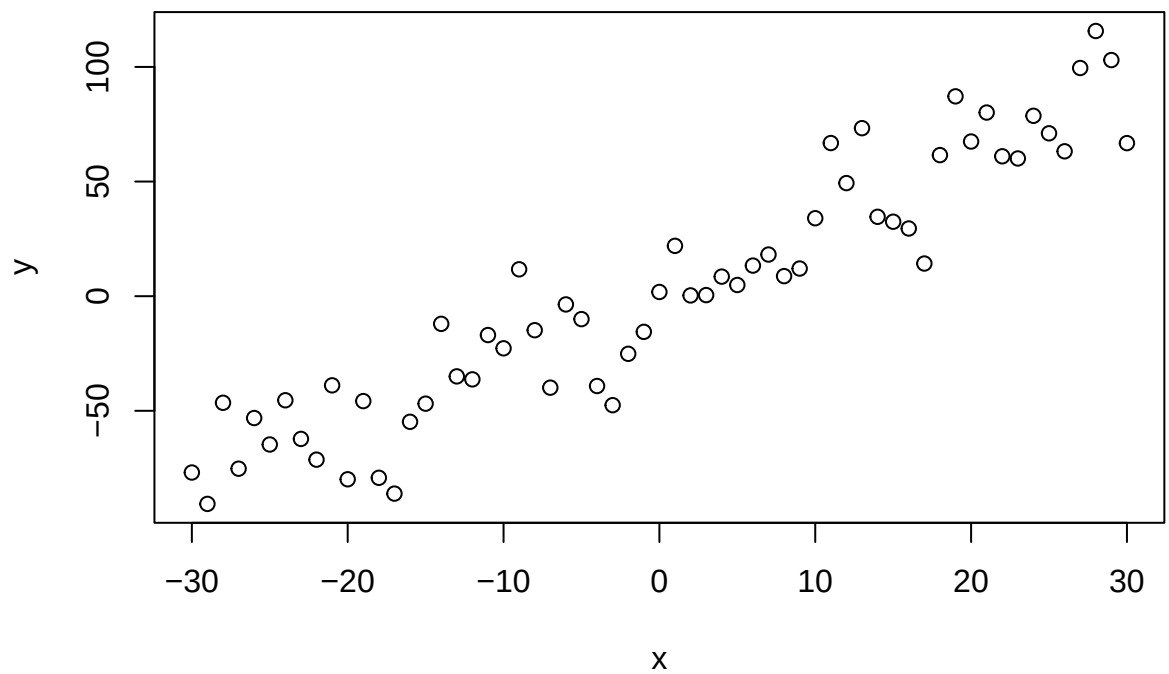
R function plot()

```
# plot of vectors
x <- 1:10
plot (x)
```




```
# Scatter plot of two vectors x and y
set.seed (13)
x <- -30:30
```

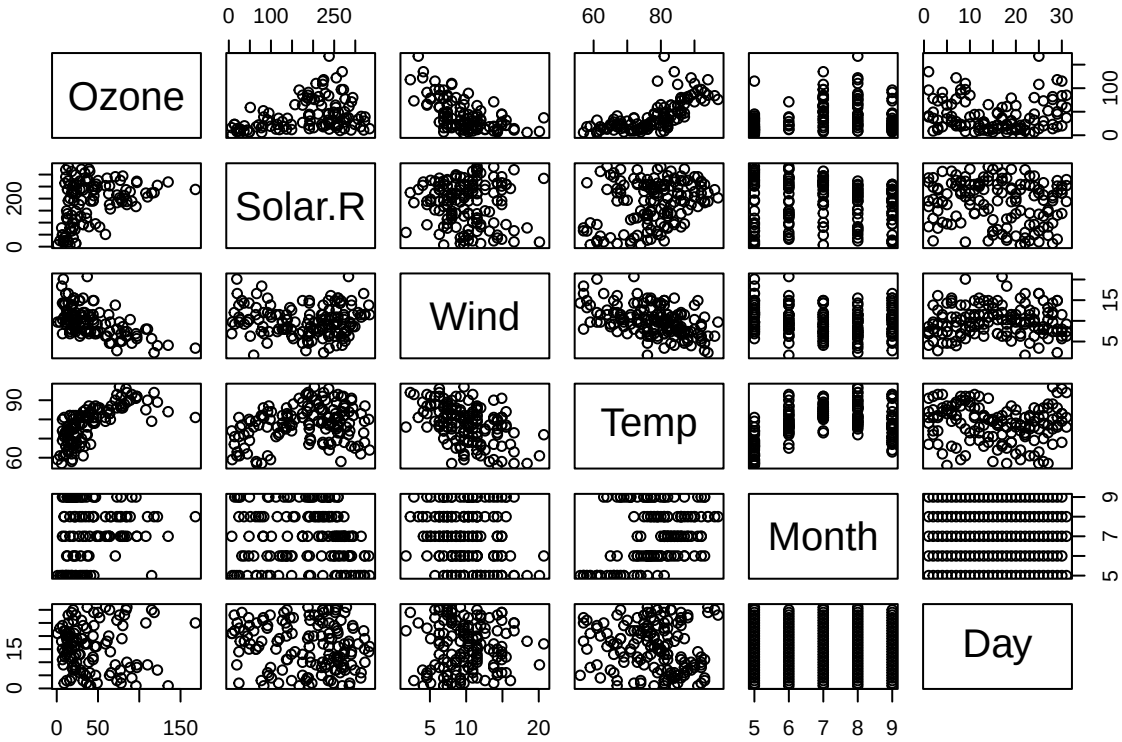
```r
y <- 3 * x + 2 + rnorm(length (x), sd = 20)
plot (x, y)
```



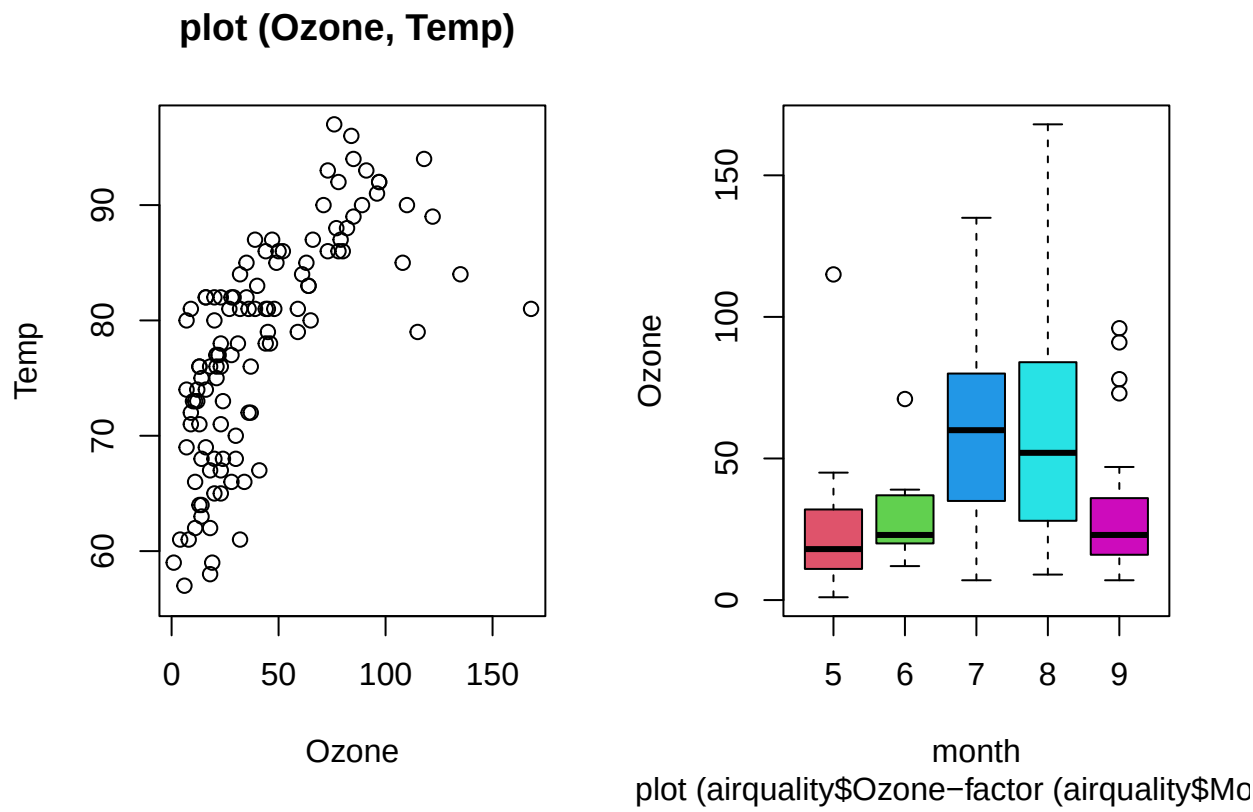

```r
# plot of data frames
airquality [1:2, ]
```

```
##   Ozone Solar.R Wind Temp Month Day
## 1    41     190  7.4   67     5   1
## 2    36     118  8.0   72     5   2
```

```r
plot (airquality) # all variables plotted against each other pairs()
```



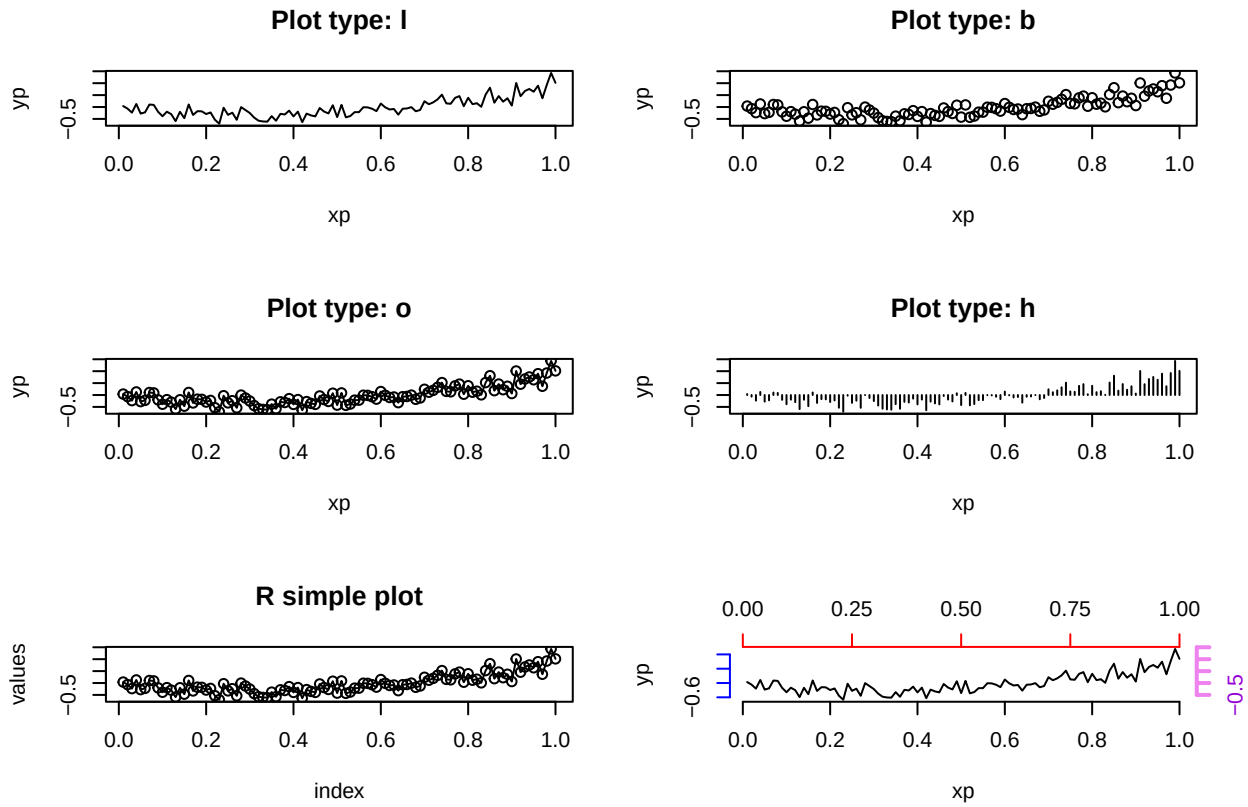

```
par (mfrow = c(1, 2))
attach (airquality)
plot (Ozone, Temp, main = "plot (Ozone, Temp) ")
plot (airquality$Ozone ~ factor (airquality$Month), col = 2:6,
sub = "plot (airquality$Ozone-factor (airquality$Month) ", ylab =
"Ozone",
xlab = "month")
```




```
detach(airquality)
```

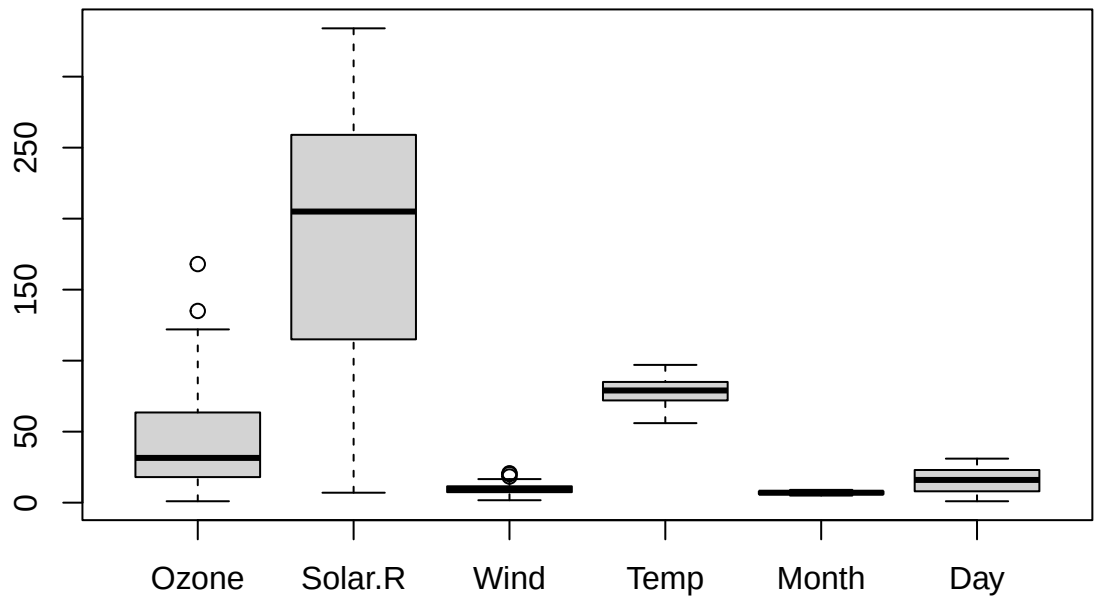
Arguments to plot

```
xp <- 1:100/100
yp <- 3 * xp^2 - 2 * xp + rnorm(100, sd = 0.2)

par (mfrow = c(3, 2))
for (i in c("l", "b", "o", "h")) plot (xp, yp, type = i, main =paste ("Plot type:",
i))

plot (xp, yp, type = "o", xlab = "index", ylab = "values",
main = "R simple plot")
plot (xp, yp, type = "l", axes = FALSE)
axis (1)
axis (2, at = c(-0.6, 0, 0.6, 1.2), col = "blue")
axis(3, at = c(0, 0.25, 0.5, 0.75, 1), col = "red")
axis (4, col = "violet", col.axis = "dark violet", lwd = 2)
```

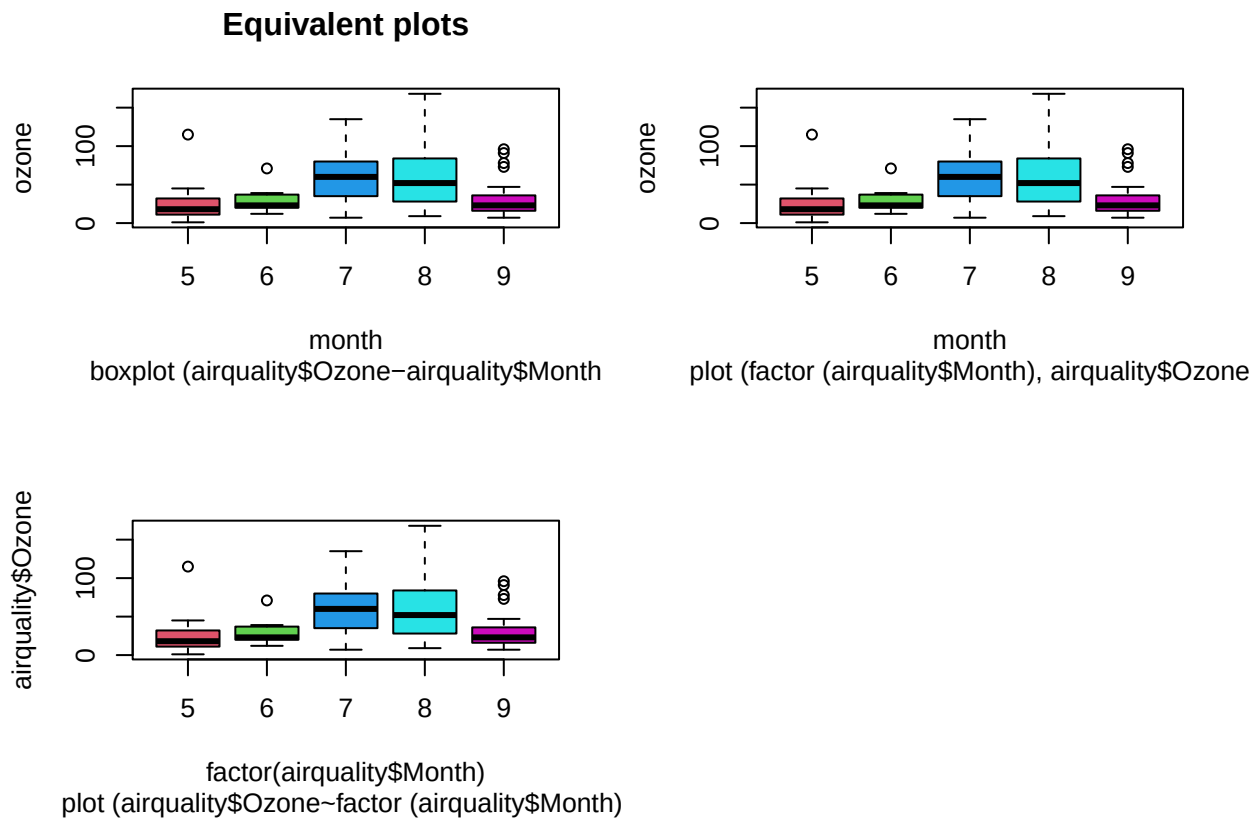


Basic graphics functions

```
boxplot(airquality)
```

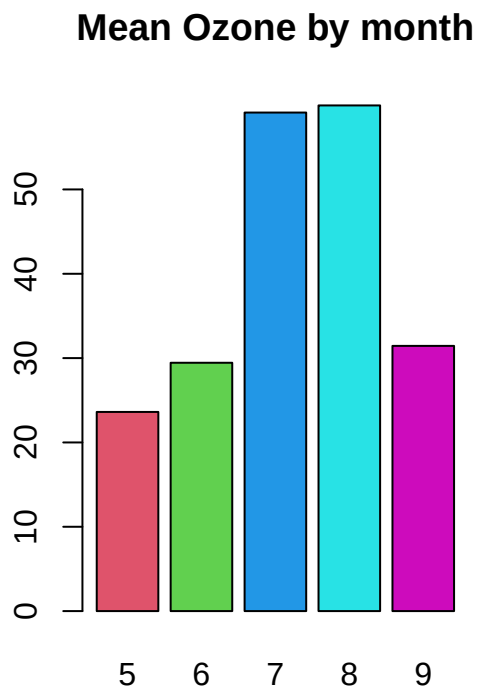



```
par (mfrow = c(2, 2))
boxplot (airquality$Ozone ~ airquality$Month, col = 2:6, xlab =
"month",
ylab = "ozone", sub =
"boxplot (airquality$Ozone-airquality$Month")
title("Equivalent plots")
plot (factor (airquality$Month), airquality$Ozone, col = 2:6,
xlab = "month", ylab = "ozone", sub =
"plot (factor (airquality$Month), airquality$Ozone")
plot (airquality$Ozone ~ factor (airquality$Month), col = 2:6,
sub = "plot (airquality$Ozone~factor (airquality$Month) ")
```



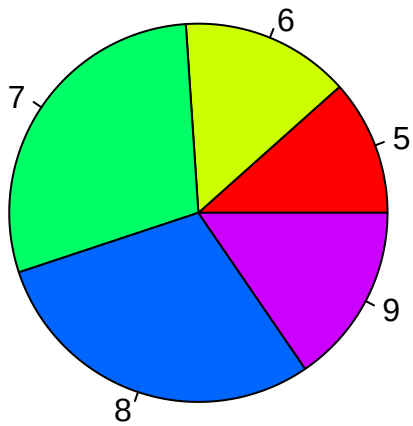
Bar plot

```
OzMonthMean <- tapply(airquality$Ozone, factor (airquality$Month),
mean, na.rm = TRUE)
par (mfrow = c(1, 2))
barplot (OzMonthMean, col = 2:6, main = "Mean Ozone by month")
```



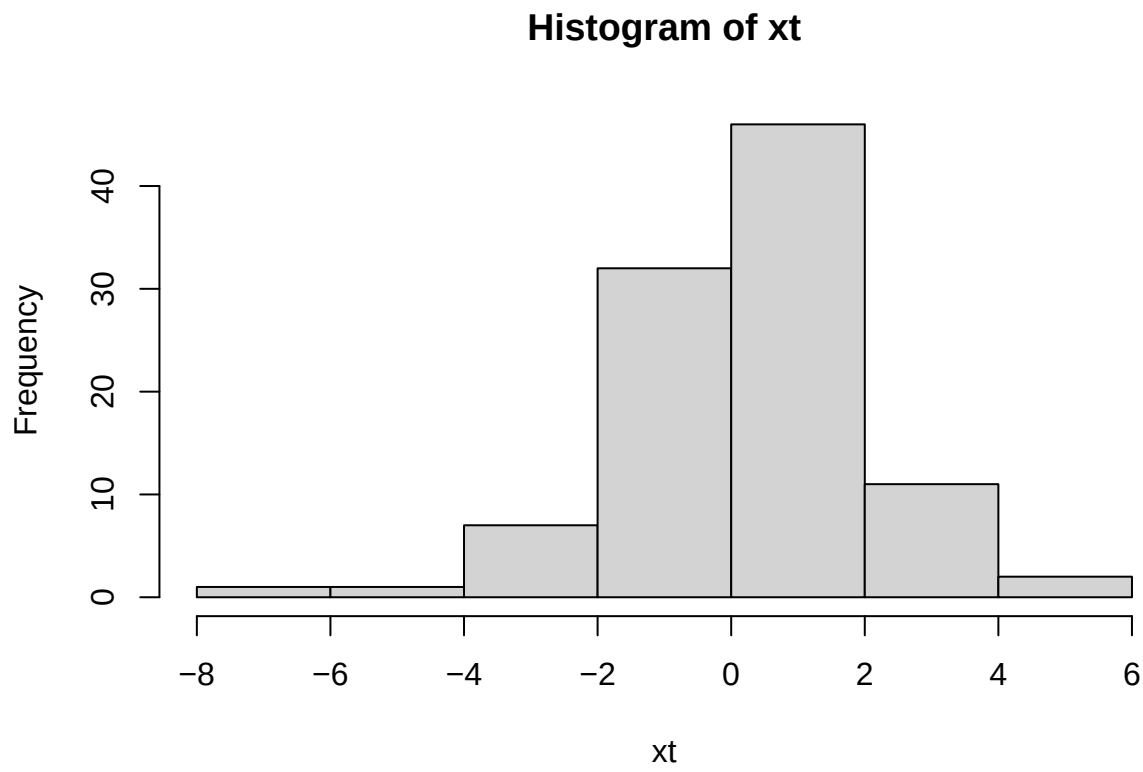
pie Charts

```
pie(OzMonthMean, col = rainbow(5))
```

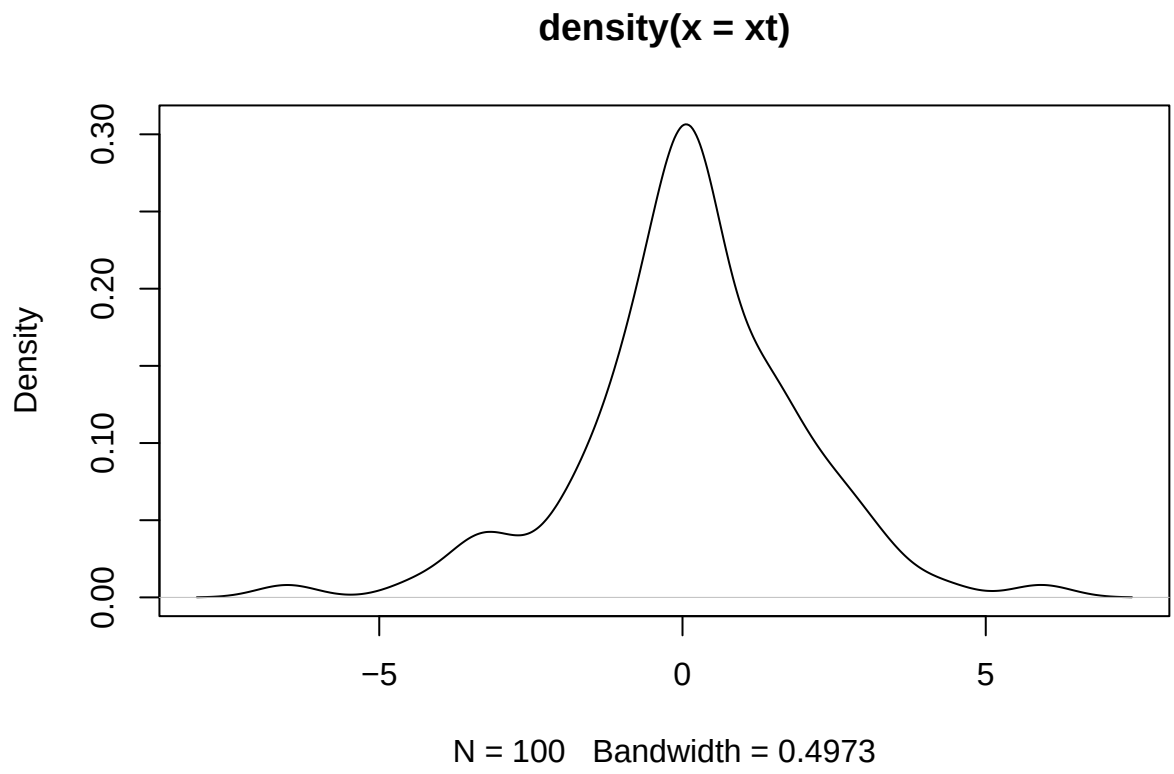


#histogram

```
xt <- rt(100, 3)
hist(xt)
```




```
plot(density(xt))
```



3D scatterplot

```
require (scatterplot3d)
```

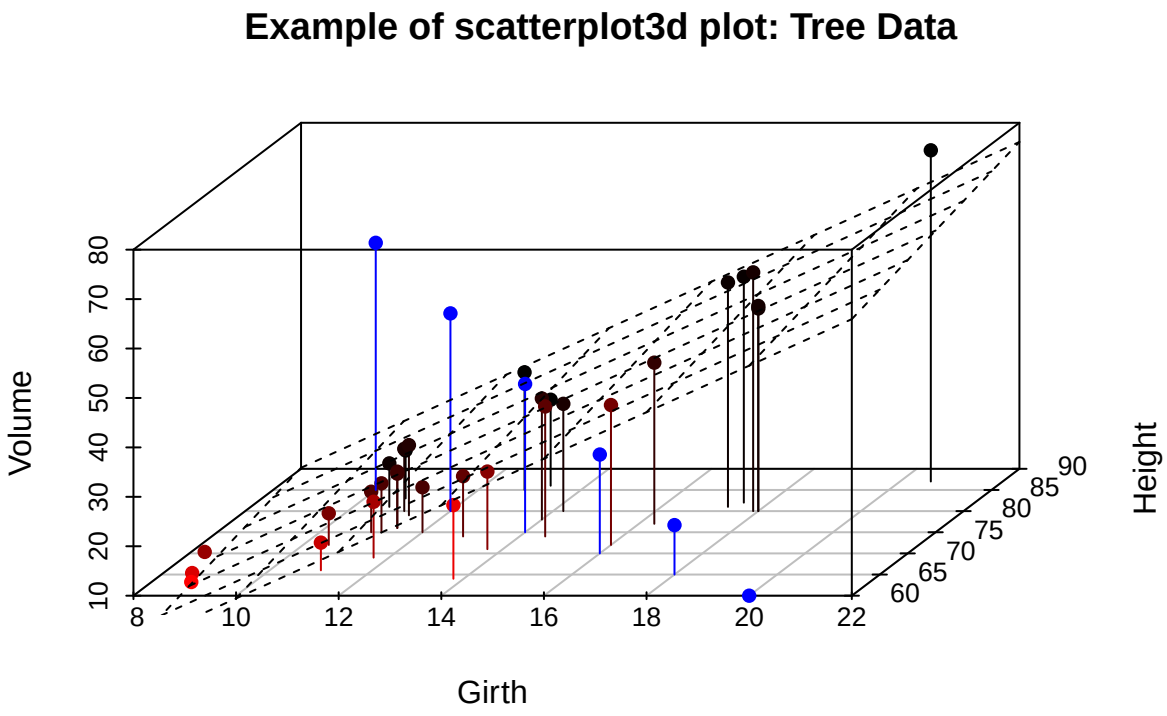
```
## Loading required package: scatterplot3d
```

```
## Loading required package: scatterplot3d
data (trees)
trees [1:2, ]
```

```
##   Girth Height Volume
## 1   8.3     70   10.3
## 2   8.6     65   10.3
```

```
s3d <- scatterplot3d(trees, type = "h", highlight.3d = TRUE, angle = 55, scale.y = 0.7,   pch = 16, mai

# Now adding some points to the 'scatterplot3d'
s3d$points3d(seq(10, 20, 2), seq(85, 60, -5), seq(60, 10,
-10), col = "blue", type = "h", pch = 16)

# Now adding a regression plane to the 'scatterplot3d'
attach (trees)
```

```r
my.lm <- lm(Volume ~ Girth + Height)
s3d$plane3d(my.lm)
```




```r
detach(trees)
```

Rvenn

```r
require (gplots)
```

```
## Loading required package: gplots

## Warning: package 'gplots' was built under R version 4.5.3

##
## ---------------------
## gplots 3.3.0 loaded:
##   * Use citation('gplots') for citation info.
##   * Homepage: https://talgalili.github.io/gplots/
##   * Report issues: https://github.com/talgalili/gplots/issues
##   * Ask questions: https://stackoverflow.com/questions/tagged/gplots
##   * Suppress this message with: suppressPackageStartupMessages(library(gplots))
## ---------------------
```

```
##
## Attaching package: 'gplots'

## The following object is masked from 'package:stats':
##
##     lowess
```

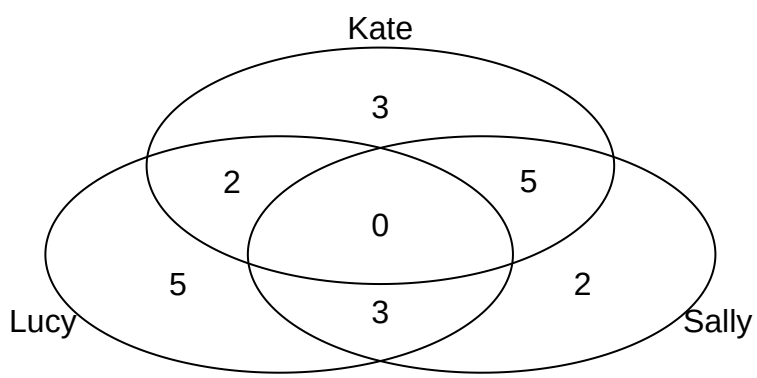
```r
sample (LETTERS, 10)
```

```
##  [1] "L" "O" "C" "H" "J" "N" "F" "Z" "I" "V"
```

```r
tt <- lapply(1:3, function(x) sample(LETTERS, 10))
names (tt) <- c("Lucy", "Sally", "Kate")
tt
```

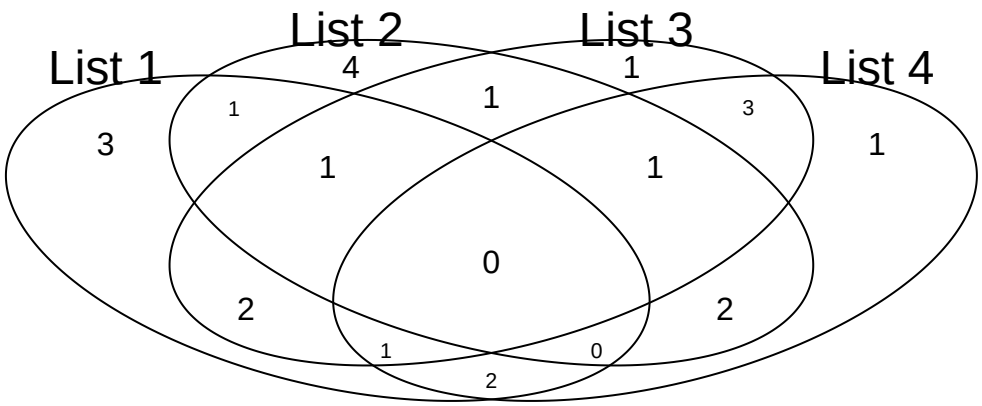
```
## $Lucy
##  [1] "L" "O" "Q" "V" "H" "U" "G" "Z" "S" "X"
##
## $Sally
##  [1] "H" "D" "A" "F" "P" "C" "R" "S" "Q" "W"
##
## $Kate
##  [1] "U" "V" "R" "N" "Y" "C" "M" "A" "F" "P"
```

```r
venn (tt)
```



plot $ intersection

```r
tt <- lapply(1:4, function(x) sample (LETTERS, 10) )
names (tt) <- paste ("List", 1:4)
venn (tt)
```



color plots

```r
require (venneuler)
```

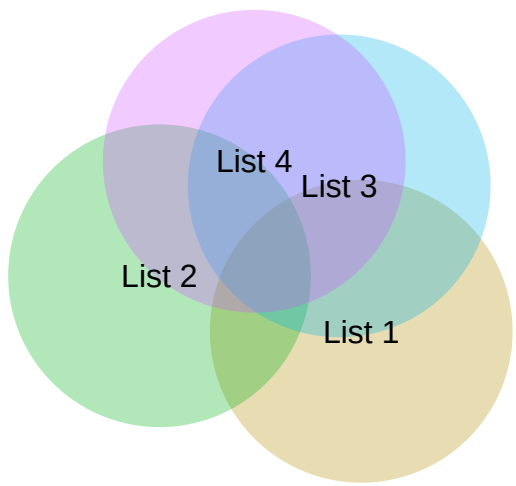
```
## Loading required package: venneuler

## Warning: package 'venneuler' was built under R version 4.5.3

## Loading required package: rJava

## Warning: package 'rJava' was built under R version 4.5.3
```

```r
## Loading required package: venneuler

IntersectMatrix <- function (tt) {
allElements <- unique(unlist (tt))
outMat <- sapply (1:length (tt), function (i) allElements %in% tt [[i]])
```

```r
rownames (outMat) <- allElements
colnames (outMat) <- names (tt)
return (outMat)
}

xx <- IntersectMatrix(tt)
print (xx [1:4, ])
```

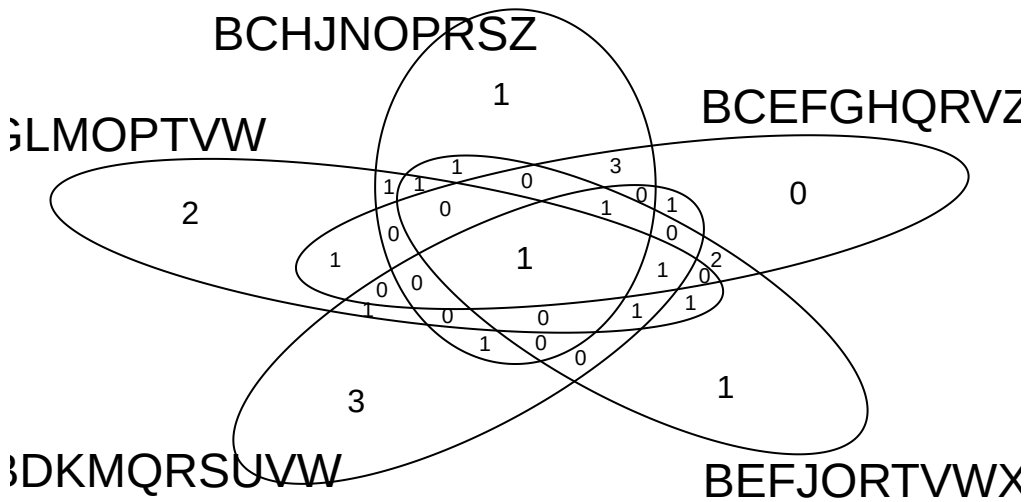
```
##   List 1 List 2 List 3 List 4
## Y   TRUE  FALSE   TRUE  FALSE
## I   TRUE  FALSE  FALSE  FALSE
## K   TRUE   TRUE   TRUE  FALSE
## G   TRUE  FALSE  FALSE  FALSE
```

```r
plot (venneuler (xx) )
```



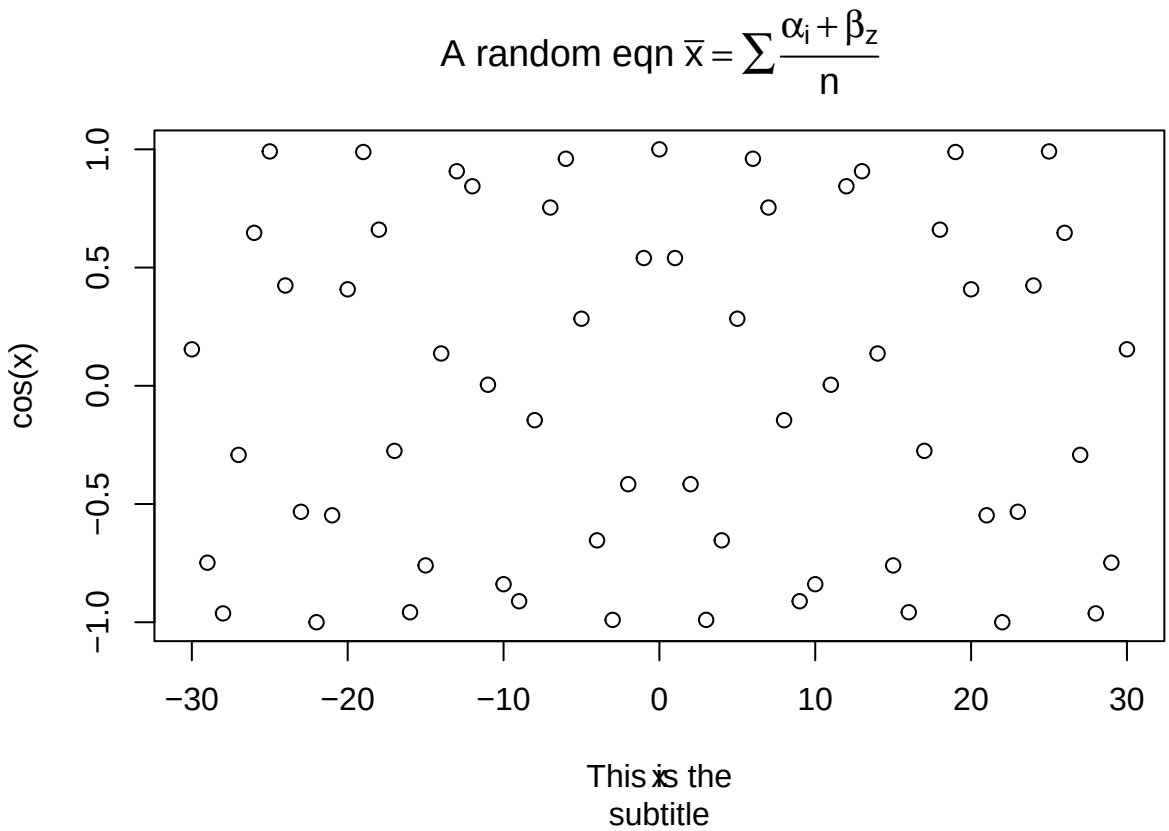
plot 5 intersections

```r
tt <- lapply(1:5, function (x) sample (LETTERS, 10))
names (tt) <- sapply(tt, function (x) paste (sort (x), collapse = "") )
venn (tt)
```



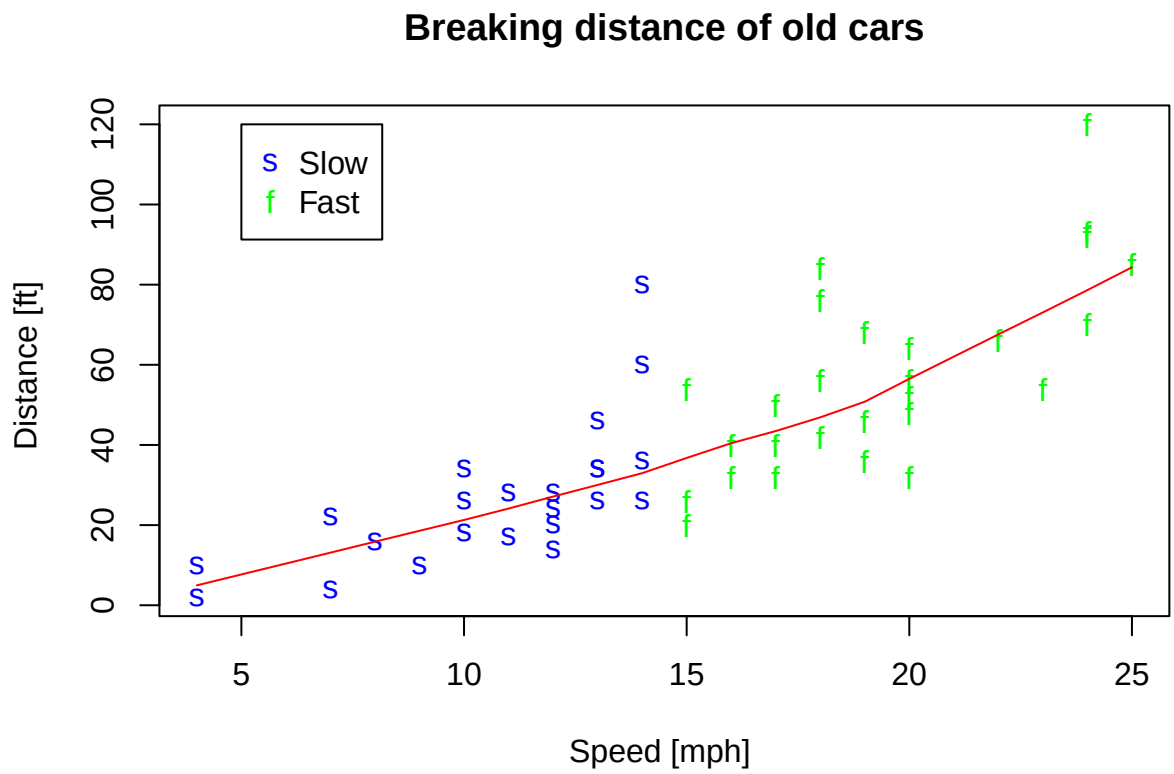
Greek Characters or function expression

```
plot (x, cos (x), main = expression (paste("A random eqn ", bar (x) ) ==
sum (frac(alpha[i] + beta[z], n) ) ), sub = "This is the
subtitle")
```



Using points lines and legends

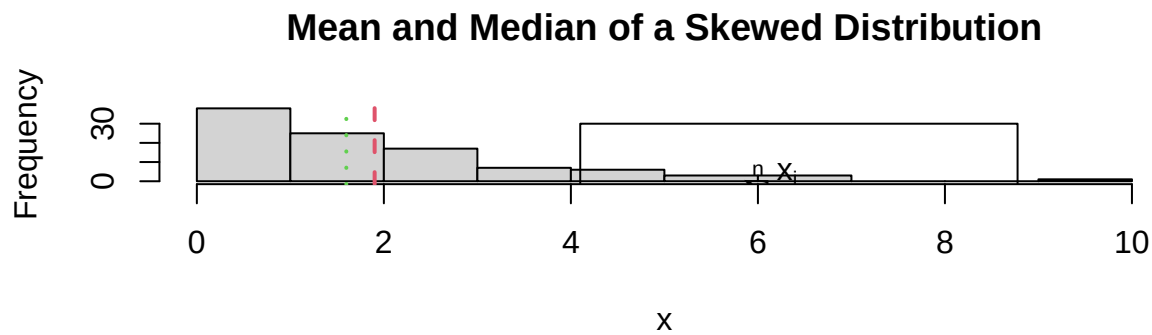
```
attach (cars)
plot (cars, type = "n", xlab = "Speed [mph] ", ylab = "Distance [ft] ")
points (speed [speed < 15], dist [speed < 15], pch = "s", col = "blue")
points (speed [speed >= 15], dist [speed >= 15], pch = "f", col = "green")
lines (lowess (cars), col = "red")
legend(5, 120, pch = c("s", "f"), col = c("blue", "green"),
legend = c("Slow", "Fast") )
title ("Breaking distance of old cars")
```

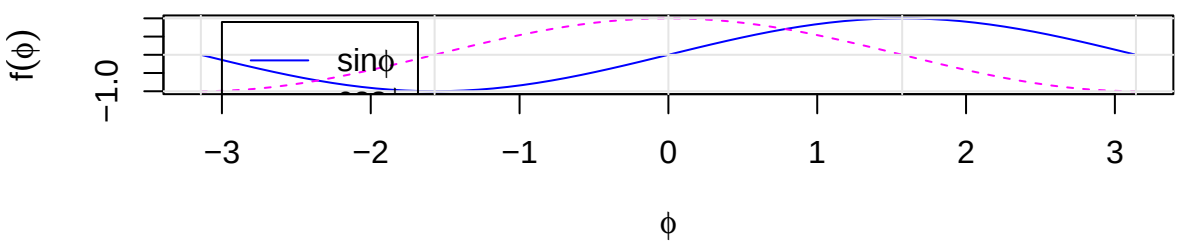



```
detach(2)
```

Formulae or greek character to a plot

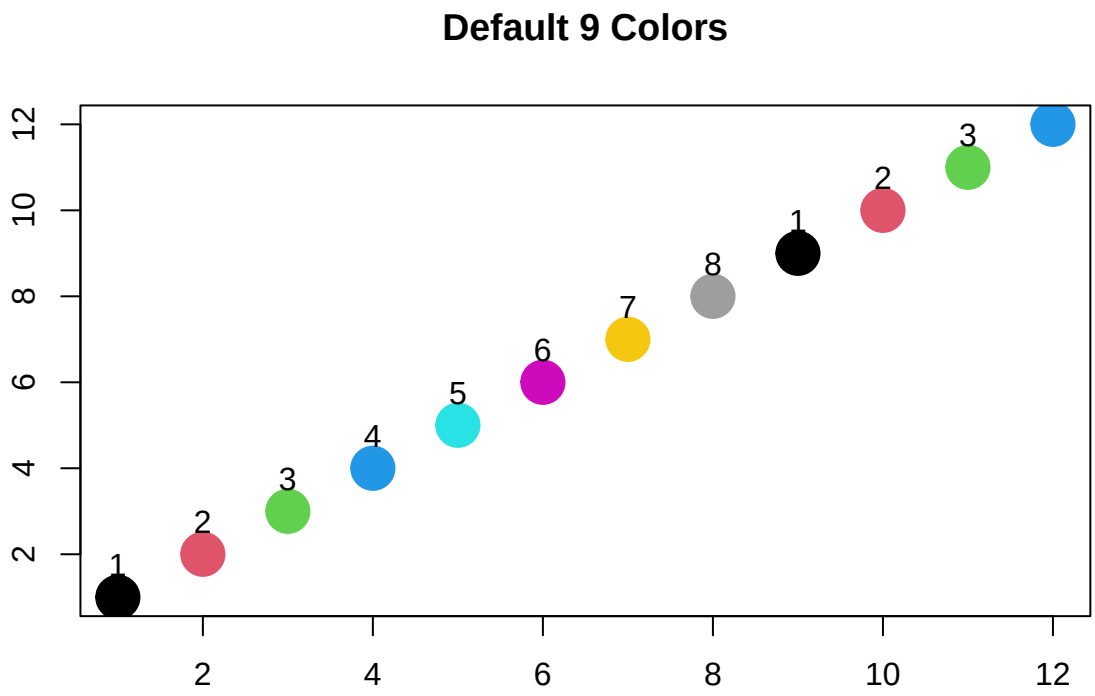
```
par (mfrow = c(2, 1) )
# Mean and Median Plot
x <- rexp (100, rate = 0.5)
hist (x, main = "Mean and Median of a Skewed Distribution")
abline (v = mean (x), col = 2, lty = 2, lwd = 2)
abline (v = median (x), col = 3, lty = 3, lwd = 2)
ex1 <- expression (bar (x) == sum(over (x[i], n) , i == 1, n),
hat (x) == median (x[i], i == 1, n) )
legend (4.1, 30, ex1, col = 2:3, lty = 2:3, lwd = 2)

x <- seq(-pi, pi, len = 65)
plot (x, sin(x), type = "l", col = "blue", xlab = expression (phi),
ylab = expression (f(phi) ))
lines (x, cos (x), col = "magenta", lty = 2)
abline (h = -1:1, v = pi/2 * (-6:6), col = "gray90")
ex2 <- expression(plain (sin) * phi, paste("cos", phi) )
legend(-3, 0.9, ex2, lty = 1:2, col = c("blue", "magenta"),
adj = c(0, 0.6))
```





R colors

```
plot (1:12, col = 1:12, main = "Default 9 Colors", ylab = "",
xlab = "", pch = 19, cex = 3)
text (1:12, c(1:12) + 0.75, c(1:8, 1:4))
```


Default 9 Colors
1
2
3
4
5
6
7
8
1
2
3
2
4
6
8
10
12
2
4
6
8
10
12